Harshith Reddy Nalla

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Education

University of South Dakota — Vermillion, SD
B.S. in Computer Science, AI Concentration | GPA: 3.6/4.0 | Expected May 2027 — *Aug 2023 – Present*

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Artificial Intelligence, Database Systems
- Data Structures & Algorithms Certificate (Java) | Dean's List

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL
ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV
Deep Learning: CNNs, Vision Transformers (ViT), Mamba (SSM), Semantic Segmentation, Fine-tuning
MLOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, ECS, SageMaker), Hugging Face Spaces, CI/CD
Backend & Frontend: FastAPI, Spring Boot, RESTful APIs, Microservices, React
Data & Evaluation: NumPy, Pandas, PostgreSQL, MySQL, Git, Dice Score, IoU, F1, ROC-AUC, Ablation Studies

Experience

AI Engineer Intern — May 2026 – Present
Sterling Trustees LLC — *Sioux Falls, SD*

- Building an **agentic RAG knowledge assistant** that answers staff questions over legal documents in Box and live Salesforce data, with citations; designed the end-to-end architecture.
- Crawled and audited **992 legal entity folders** in Box (approx. 390K PDFs), producing a structured manifest and an ingestion governance plan.
- Built a **Python script** that extracts account numbers and dates from custodial statement PDFs and auto-files them into the correct Box folders via the **Box API**, replacing a manual filing process.
- Scoped **Salesforce automations** to replace manual data entry, including a **Box AI metadata extraction** to Salesforce sync pipeline.

AI Research Assistant — Jun 2025 – Present
University of South Dakota, Computer Science Department — *Vermillion, SD*

- Trained **3D medical image segmentation** models for liver tumor and colorectal polyp detection using PyTorch CNNs, Vision Transformers, and Mamba (SSM) architectures.
- Conducted systematic **ablation studies** across CNN, Transformer, and Mamba architectures, evaluating Dice, IoU, sensitivity, and specificity; findings contributed to **1 peer-reviewed publication** (AAAI 2026 workshop).
- Built end-to-end **MLOps pipelines**: packaged models into **FastAPI + Docker** microservices deployed on Hugging Face Spaces for real-time inference.

Entrepreneurial Lead – DentiMap AI — Oct 2025 – Nov 2025
NSF I-Corps Program, National Science Foundation — *Vermillion, SD*

- Led **customer discovery** for an AI dental diagnostics venture, conducting **20+ structured interviews** with dentists and clinic owners to validate product-market fit.

Projects

Polyp Detection | *PyTorch, CNNs, Vision Transformers, Semantic Segmentation, FastAPI, React* — 2025

- Trained a colonoscopy polyp segmentation model achieving **86% Dice score**; deployed a **FastAPI inference pipeline** on Hugging Face Spaces with a React frontend.

DentiMap | *PyTorch, CNNs, Spring Boot, FastAPI, Docker, React, TypeScript* — 2025

- Built a full-stack AI dental diagnostics platform achieving **87% caries detection accuracy** with **sub-1s inference latency**; deployed on Hugging Face Spaces with a React/TypeScript frontend.

Publications

When CNNs Outperform Transformers and Mambas – *AAAI 2026 Workshop* — 2025

- A. Ghimire, J. Zeng, R. Paudel, N. K. Tomar, D. R. Nayak, **H. R. Nalla**, V. Jha, G. Reynolds, D. Jha | *AAAI 2026, Singapore*